

CERVICAL SPECIAL TESTS

1. Spurling's Test (Foraminal Compression Test)

PURPOSE

To assess cervical nerve root compression (cervical radiculopathy).

PROCEDURE

- Patient sits comfortably.
- Extend, laterally flex and rotate the head towards the affected side.
- Apply gentle downward compression on the head.

POSITIVE SIGN

Radiating pain, numbness or tingling into the upper limb.

INDICATES

- Cervical nerve root compression.
- Cervical radiculopathy.

2. Cervical Distraction Test

PURPOSE

To assess cervical nerve root compression.

PROCEDURE

- Patient lies supine.
- Therapist gently grasps the head and applies an upward traction force.

POSITIVE SIGN

Relief of neck or arm pain.

INDICATES

- Cervical radiculopathy.
- Foraminal compression.

3. Upper Limb Tension Test (ULTT)

PURPOSE

To assess neural tissue mobility and detect peripheral nerve or cervical nerve root involvement.

TYPES OF ULTT

ULTT	Nerve Biased	Common Indication
ULTT 1 (ULTT A)	Median nerve	Median nerve tension, Cervical radiculopathy
ULTT 2A	Median nerve (different bias)	Median nerve involvement
ULTT 2B	Radial nerve	Radial nerve entrapment/tension
ULTT 3	Ulnar nerve	Ulnar nerve entrapment/tension

★ **Exam Point: ULTT 1 is the most sensitive test for cervical radiculopathy.**

ULTT 1 (MEDIAN NERVE BIAS)

Procedure

- Depress the shoulder.
- Abduct the shoulder to 110°.
- Supinate the forearm.
- Extend the wrist and fingers.
- Externally rotate the shoulder.
- Slowly extend the elbow.
- Finally, side flex the neck away if required.

Positive Sign

- Pain, tingling or numbness along the median nerve distribution.
- Difference in symptoms compared to the opposite side.

Indicates

- Median nerve tension.
- Cervical radiculopathy.

ULTT 2A (MEDIAN NERVE BIAS)

Procedure

- Shoulder depression.
- Elbow extension.
- External rotation of shoulder.
- Forearm supination.
- Wrist and finger extension.

Positive Sign

Reproduction of median nerve symptoms.

Indicates

Median nerve irritation.

ULTT 2B (RADIAL NERVE BIAS)

Procedure

- Shoulder depression.
- Internal rotation of shoulder.
- Elbow extension.
- Forearm pronation.
- Wrist, finger and thumb flexion with ulnar deviation.

Positive Sign

Pain or tingling along the dorsolateral forearm and hand.

Indicates

Radial nerve tension.

ULTT 3 (ULNAR NERVE BIAS)

Procedure

- Shoulder depression.
- Shoulder abduction and external rotation.
- Elbow flexion.
- Forearm pronation.
- Wrist extension with radial deviation.
- Finger extension.

Positive Sign

Pain or tingling along the medial forearm and little finger.

Indicates

Ulnar nerve tension.

4. Vertebral Artery Test

PURPOSE

To assess vertebrobasilar insufficiency before cervical mobilization or manipulation.

PROCEDURE

- Patient lies supine.
- Therapist extends, rotates and laterally flexes the neck.

- Hold for about 30 seconds while observing symptoms.

POSITIVE SIGN

- Dizziness
- Vertigo
- Nystagmus
- Blurred vision
- Nausea

INDICATES

Vertebrobasilar insufficiency.

★ **Precaution: Stop the test immediately if symptoms appear.**

★ Revision Table – Cervical Spine Special Tests

(Reference: David J. Magee – Orthopedic Physical Assessment)

Test	Purpose	Positive Sign	Indicates
Spurling's Test (Foraminal Compression Test)	Assess cervical nerve root compression	Radiating pain, numbness or tingling into the upper limb	Cervical radiculopathy / Foraminal compression
Cervical Distraction Test	Assess cervical nerve root compression	Relief of neck or arm pain	Cervical radiculopathy / Foraminal compression
Upper Limb Tension Test (ULTT 1)	Assess median nerve tension	Pain/tingling along median nerve distribution	Median nerve tension, Cervical radiculopathy
Upper Limb Tension Test (ULTT 2A)	Assess median nerve (alternate bias)	Reproduction of median nerve symptoms	Median nerve involvement
Upper Limb Tension Test (ULTT 2B)	Assess radial nerve tension	Pain/tingling over dorsolateral forearm and hand	Radial nerve involvement
Upper Limb Tension Test (ULTT 3)	Assess ulnar nerve tension	Pain/tingling over medial forearm and little finger	Ulnar nerve involvement
Vertebral Artery Test	Assess vertebrobasilar insufficiency	Dizziness, vertigo, nystagmus, blurred vision, nausea	Vertebrobasilar insufficiency

SPECIAL EXAMINATION TESTS

1. DIZZINESS TEST ★★★★★

PURPOSE

To assess for cervical vascular compromise/vertebrobasilar insufficiency that may produce dizziness during cervical movement.

PROCEDURE

- Patient is positioned sitting.
- Examiner stabilizes the patient's head.
- Slowly place the cervical spine into extension and rotation.
- Observe the patient for symptoms.

POSITIVE SIGN

- Dizziness
- Vertigo
- Nausea
- Visual disturbance
- Nystagmus

INDICATES

Possible vertebrobasilar insufficiency and requires further medical evaluation.

2. SHOULDER DEPRESSION TEST ★★★★★

PURPOSE

To assess nerve-root/nerve tissue irritation or shortening of neural structures in the cervical and upper-limb region.

PROCEDURE

- Patient is sitting.
- Examiner stabilizes/depresses the shoulder on the tested side.
- Passively side-flex the patient's neck away from the tested side.
- The position places tension on the cervical nerve structures.

POSITIVE SIGN

- Reproduction of the patient's radiating pain/paresthesia into the upper limb.

INDICATES

Possible cervical nerve-root or brachial plexus involvement.

3. PINCH GRIP TEST ★★★★★

PURPOSE

To assess the function of the anterior interosseous nerve (AIN) and flexor pollicis longus/index FDP function.

PROCEDURE

- Ask the patient to make a pinch/"OK" sign by touching the tip of the thumb to the tip of the index finger.
- Observe the shape of the circle formed.

POSITIVE SIGN

- Patient cannot make a proper round "OK" circle.
- Instead, the pinch becomes flattened/pulp-to-pulp.

INDICATES

Anterior interosseous nerve palsy/dysfunction.

4. CARPAL COMPRESSION TEST ★★★★★

PURPOSE

To assess for carpal tunnel syndrome due to median nerve compression.

PROCEDURE

- Patient's wrist is in a neutral position.
- Examiner applies direct pressure over the carpal tunnel with the thumbs/fingers.
- Maintain pressure for approximately 30 seconds.

POSITIVE SIGN

- Reproduction of:
 - Pain
 - Tingling
 - Numbness
 - Paresthesia in the median nerve distribution

INDICATES

Carpal Tunnel Syndrome / median nerve compression.

5. HAND ELEVATION TEST ★★★★★

PURPOSE

To assess for carpal tunnel syndrome/median nerve compression.

PROCEDURE

- Patient raises both hands above the head.
- Maintain the elevated position for approximately 1–2 minutes.
- Ask the patient about symptoms.

POSITIVE SIGN

- Reproduction of pain, numbness or tingling in the median nerve distribution.

INDICATES

Possible Carpal Tunnel Syndrome.

6. FROMENT'S SIGN ★★★★★

PURPOSE

To assess the ulnar nerve, particularly the function of adductor pollicis.

PROCEDURE

- Ask the patient to hold a piece of paper between the thumb and index finger.
- Examiner attempts to pull the paper away.

POSITIVE SIGN

Patient compensates by:

- Flexing the IP joint of the thumb using flexor pollicis longus.

INDICATES

Ulnar nerve palsy/weakness of adductor pollicis.

★ QUICK REVISION TABLE

Test	Purpose	Positive Finding	Indicates
Dizziness Test	Cervical vascular assessment	Dizziness/vertigo/visual symptoms	Possible vertebrobasilar insufficiency
Shoulder Depression	Neural tissue/nerve-root assessment	Radiating pain/paresthesia	Cervical nerve-root/brachial plexus involvement
Pinch Grip Test	AIN function	Abnormal/flattened OK sign	AIN dysfunction
Carpal Compression	Median nerve	Paresthesia/pain	Carpal tunnel syndrome
Hand Elevation	Median nerve	Reproduction of symptoms	Carpal tunnel syndrome
Froment's Sign	Ulnar nerve/adductor pollicis	Thumb IP flexion	Ulnar nerve palsy